
DECOMPOSING INFORMATION CONTENT IN POSTERIORs: A FLOW-MATCHING PERSPECTIVE

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ABSTRACT

We propose a novel information-theoretic framework to evaluate inversion methods by leveraging the additive decomposition of the Conditional Flow Matching (CFM) operator. By partitioning the vector field into deterministic, Gaussian-equivalent, and non-linear structural components, we provide a diagnostic tool to quantify the information recovery capacity of arbitrary inversion algorithms.

1 Introduction

Current inversion methods often struggle to resolve non-Gaussian features in Earth System dynamics. We demonstrate that the CFM operator $\Psi(X_\tau, Z, \tau)$ provides a canonical decomposition that maps directly to physical modes of information[cite: 96, 110].

In the notes below, we consider random processes $\mathbf{X}_1$ and $\mathbf{Z}$ taking values in $\mathcal{X}$ and $\mathcal{Z}$. This accounts for a wide range of configurations:

- **State $\mathbf{X}_1$:** state, state trajectories, model parameters;
- **Conditioning $\mathbf{Z}$:** previous states, observations, forcings, model parameters....

2 Theoretical Background

The flow-matching framework characterizes the transformation from a latent Gaussian prior to a target posterior. To define this process, we consider random processes $\mathbf{X}_1$ and $\mathbf{Z}$ taking values in $\mathcal{X}$ and $\mathcal{Z}$, respectively[cite: 14, 23].

2.1 The Gaussian Probability Path

The latent state $\mathbf{X}_\tau$ at any time $\tau \in [0, 1]$ is defined by the interpolation between the target state $\mathbf{X}_1$ and standard Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbb{I})$ [cite: 55]:

$$\mathbf{X}_\tau = \beta(\tau)\mathbf{X}_1 + \alpha(\tau)\epsilon \quad (1)$$

where $\alpha(\tau)$ and $\beta(\tau)$ are scalar functions satisfying the boundary conditions $\beta(1) = \alpha(0) = 1$ and $\beta(0) = \alpha(1) = 0$ [cite: 14]. This path ensures the latent state at $\tau = 0$ is purely Gaussian[cite: 15] conditionally to the true state.

2.2 The Flow-Matching Operator

The evolution of the latent state is governed by the following ODE system[cite: 15]:

$$\begin{cases} \mathbf{X}_0 \sim \mathcal{N}(0, \mathbb{I}) \\ \frac{d\mathbf{X}_\tau}{d\tau} = \frac{\dot{\alpha}_\tau}{\alpha_\tau} \mathbf{X}_\tau + \left(\dot{\beta}_\tau - \beta_\tau \frac{\dot{\alpha}_\tau}{\alpha_\tau} \right) \Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) \end{cases} \quad (2)$$

The operator Ψ is the core of the inversion framework. Within the flow matching framework, this operator is given by the conditional expectation of the target given the current latent state and context [1, 2]:

$$\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \mathbb{E}[\mathbf{X}_1 | \mathbf{X}_\tau, \mathbf{Z}] \quad (3)$$

More specifically, the following theorem holds according to the uniqueness of the operator Ψ under the considered linear flow matching path (1)

Theorem 1. For given scalar functions $\alpha(\cdot)$ and $\beta(\cdot)$ such that $\beta(1) = \alpha(0) = 1$, $\beta(0) = \alpha(1) = 0$, and $\alpha(\tau) \geq 0, \beta(\tau) \geq 0$, let $\mathbf{X}_1$ and $\mathbf{Z}$ be random processes[cite: 14]. Let $\mathbf{X}_\tau | \mathbf{X}_1$ follow a Gaussian distribution with mean $\beta(\tau)\mathbf{X}_1$ and covariance $\alpha(\tau)^2 \mathbb{I}$ [cite: 14]. The solutions of the ODE system:

$$\begin{cases} \mathbf{X}_0 \sim \mathcal{N}(0, \mathbb{I}) \\ \frac{d\mathbf{X}_\tau}{d\tau} = \frac{\dot{\alpha}(\tau)}{\alpha(\tau)}\mathbf{X}_\tau + (\dot{\beta}(\tau) - \beta(\tau)\frac{\dot{\alpha}(\tau)}{\alpha(\tau)})\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) \end{cases} \quad (4)$$

sample the posterior $p(\mathbf{X}_\tau | \mathbf{Z})$ if and only if the operator Ψ satisfies[cite: 15, 17]:

$$\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \mathbb{E}[\mathbf{X}_1 | \mathbf{X}_\tau, \mathbf{Z}] \quad (5)$$

This equivalence follows from the uniqueness of the flow matching velocity field defined by the linear flow matching path between the latent Gaussian prior and the targeted posterior.

2.3 Optimal Transport and Convergence

NB: probably not relevant here. BUT to likely useful to interpret the flow matching operator as multi-scale analysis.

Under Optimal Transport (OT) scheduling constraints, where $\alpha(\tau) = 1 - \tau$ and $\beta(\tau) = \tau$, the system achieves a uniform asymptotic convergence rate of $\mathcal{O}((1 - \tau)^2)$ for all frequencies ω in terms of Mean Squared Error (MSE)[cite: 78, 84]. The error manifold for any frequency is given by[cite: 74]:

$$MSE(\omega, \tau) = \frac{S(\omega)\alpha^2(\tau)}{\beta^2(\tau)S(\omega) + \alpha^2(\tau)} \quad (6)$$

where $S(\omega) = \mathbb{E}[|\mathbf{X}_1(\omega)|^2]$ represents the expected spectral power of the target state[cite: 53]. This confirms that the flow-matching operator acts as an inherent multi-scale filter, prioritizing large-scale structures ($S(\omega) \rightarrow \infty$) over fine-scale structures ($S(\omega) \rightarrow 0$)[cite: 93].

3 The Information-Theoretic Decomposition

3.1 Additive Decomposition of Information

Given the validity of Theorem 1, we can partition the velocity field into three physical counterparts[cite: 96]:

$$\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \Psi_{mean}(\mathbf{Z}) + \Psi_G(\mathbf{X}_\tau, \mathbf{Z}, \tau) + \Psi_{NG}(\mathbf{X}_\tau, \mathbf{Z}, \tau) \quad (7)$$

where each term corresponds to the deterministic background, the Gaussian-equivalent update, and the non-linear structural residual respectively[cite: 96, 110]:

- **Deterministic Mean (Ψ_{mean}):** Representing the deterministic background state, this term is defined as $\Psi_{mean}(\mathbf{Z}) = \mathbb{E}[\boldsymbol{\mu}_1 | \mathbf{Z} | \mathbf{X}_0, \mathbf{Z}] = \mathbb{E}[\boldsymbol{\mu}_1 | \mathbf{Z} | \mathbf{Z}] = \boldsymbol{\mu}_1 | \mathbf{Z}$ [cite: 113]. It captures the unperturbed trajectory driven by boundary conditions[cite: 115].
- **Gaussian Anomaly (Ψ_G):** This component captures the second-order moments of the posterior spread[cite: 104]. It resolves to an optimal linear update via the time-dependent gain matrix $\mathbf{K}_\tau = (\alpha^2(\tau)\mathbb{I} + \beta^2(\tau)\Sigma_{1|\mathbf{Z}}^{-1})^{-1}$ [cite: 141, 149] and $\Psi_G(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \mathbf{K}_\tau(\mathbf{X}_\tau - \beta(\tau)\Psi_{mean}(\mathbf{Z}))$
- **Non-Gaussian Structural Residual (Ψ_{NG}):** Using Tweedie's Formula, this residual captures non-linear structural errors, multi-modality, and physical boundaries[cite: 105, 123]. It is defined as:

$$\Psi_{NG}(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \frac{\alpha^2(\tau)}{\beta(\tau)} \mathcal{R}_{score}(\mathbf{X}_\tau, \mathbf{Z}, \tau) \quad (8)$$

3.2 Flow-Matching-based Similarity Measure

To quantify the divergence between posteriors $p(\mathbf{X}_1|\mathbf{Z})$ and $p(\tilde{\mathbf{X}}_1|\mathbf{Z})$, we define the divergence $\mathcal{S}_{\mathbf{Z},p}^*(\mathbf{X}_1, \tilde{\mathbf{X}}_1)$ based on the associated flow-matching operators Ψ and $\tilde{\Psi}$:

$$\mathcal{S}_{\mathbf{Z},p}^*(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = \int_{[0,1]} \mathbb{E}[\|\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) - \tilde{\Psi}(\mathbf{X}_\tau, \mathbf{Z}, \tau)\|_p^p] d\tau \quad (9)$$

From this, we derive the symmetric similarity measure $\mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \tilde{\mathbf{X}}_1)$ [cite: 30]:

$$\mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = \mathcal{S}_{\mathbf{Z},p}^*(\mathbf{X}_1, \tilde{\mathbf{X}}_1) + \mathcal{S}_{\mathbf{Z},p}^*(\tilde{\mathbf{X}}_1, \mathbf{X}_1) \quad (10)$$

Properties: The measure $\mathcal{S}_p$ possesses several key properties that make it a robust diagnostic tool[cite: 35]:

- **Norm Space:** It defines a norm in the space of posteriors, specifically $\mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = \mathcal{S}_p(\mathbf{X}_1 - \tilde{\mathbf{X}}_1|\mathbf{Z}, 0)$ [cite: 35, 36].
- **Identity of Indiscernibles:** $\mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = 0$ implies $\mathbf{X}_1|\mathbf{Z} = \tilde{\mathbf{X}}_1|\mathbf{Z}$ [cite: 37].
- **Strictly Proper Scoring Rule:** As the measure relates to L_p norms in the velocity space, it satisfies $\mathcal{S}_{\mathbf{Z},p}(\tilde{\mathbf{X}}_1, \mathbf{X}_1) \geq \mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \mathbf{X}_1) \geq 0$, ensuring that the optimal estimator recovers the true posterior[cite: 45, 46].
- **Linearity:** The flow-matching operator is linear, since the flow matching of random process $a\mathbf{X}_1 + b\tilde{\mathbf{X}}_1|\mathbf{Z}$ for any scalars a and b is given by $a\Psi + b\tilde{\Psi}$ [cite: 26, 27].

Similarity Measure according to information decomposition (7): To evaluate the similarity between an inversion method sampling a distribution $\tilde{\mathbf{X}}_1|\mathbf{Z}$ and the target posterior $\mathbf{X}_1|\mathbf{Z}$ while respecting the underlying physical dynamics, we define the anomaly-aware symmetric similarity measure $\mathcal{S}_{\mathbf{Z},p}^{G,NG}$ based on the additive decomposition of the flow-matching operator:

$$\mathcal{S}_{\mathbf{Z},p}^{G,NG}(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = \mathcal{S}_{\mathbf{Z},p}(\Psi_{mean}, \tilde{\Psi}_{mean}) + \mathcal{S}_{\mathbf{Z},p}(\Psi_G, \tilde{\Psi}_G) + \mathcal{S}_{\mathbf{Z},p}(\Psi_{NG}, \tilde{\Psi}_{NG}) \quad (11)$$

The decomposition allows for a granular assessment of information recovery:

- **Deterministic Mean Divergence:** The first term, $\mathcal{S}_{\mathbf{Z},p}(\Psi_{mean}, \tilde{\Psi}_{mean})$, isolates errors in the deterministic background state[cite: 43]. This identifies if the method fails to integrate the context $\mathbf{Z}$ into the mean drift[cite: 43].
- **Covariance-based Divergence:** The second term, $\mathcal{S}_{\mathbf{Z},p}(\Psi_G, \tilde{\Psi}_G)$, measures discrepancy in the covariance structure of two distributions.
- **Non-Gaussian-based Divergence:** $\mathcal{S}_{\mathbf{Z},p}(\Psi_{NG}, \tilde{\Psi}_{NG})$, measures discrepancy in the non-Gaussian anomaly space[cite: 43].

Proper Scoring Rule Property: Because $\mathcal{S}_p$ and $\mathcal{S}_p^A$ relate to L_p norms in the flow-matching velocity space, they define strictly proper scoring rules[cite: 45]. Specifically, for any candidate inversion $\tilde{\mathbf{X}}_1$:

$$\mathcal{S}_{\mathbf{Z},p}(\tilde{\mathbf{X}}_1, \mathbf{X}_1) \geq \mathcal{S}_{\mathbf{Z},p}(\mathbf{X}_1, \mathbf{X}_1) \geq 0 \quad (12)$$

These measures are minimized to zero if and only if the inversion method recovers the true conditional distribution, i.e., $\tilde{\mathbf{X}}_1|\mathbf{Z} = \mathbf{X}_1|\mathbf{Z}$ [cite: 46].

4 Relationship to Fisher Information and the Information Gap

The diagnostic power of our decomposition relies on the relationship between the flow-matching operator Ψ and the score function $\mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \nabla_{\mathbf{X}_\tau} \log p(\mathbf{X}_\tau|\mathbf{Z})$.

4.1 Score-Operator Equivalence

By Tweedie's Formula, the optimal flow-matching operator is linked to the score function:

$$\Psi(\mathbf{X}_\tau, \mathbf{Z}, \tau) = \frac{\mathbf{X}_\tau}{\beta(\tau)} + \frac{\alpha^2(\tau)}{\beta(\tau)} \mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau) \quad (13)$$

This confirms that any inversion method approximating Ψ is implicitly performing score-based estimation[cite: 136, 137].

4.2 Similarity Measure as Fisher Divergence

The Fisher Information Matrix (FIM), $\mathcal{I}(\tau)$, characterizes the curvature of the log-posterior:

$$\mathcal{I}(\tau) = \mathbb{E}_{\mathbf{X}_\tau, \mathbf{Z}} [\mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau) \mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau)^\top] \quad (14)$$

For $p = 2$, our similarity measure $\mathcal{S}_{\mathbf{Z},2}^*$ relates to the Fisher divergence:

$$\mathcal{S}_{\mathbf{Z},2}^*(\mathbf{X}_1, \tilde{\mathbf{X}}_1) = \int_{[0,1]} \frac{\alpha^4(\tau)}{\beta^2(\tau)} \mathbb{E}_{\mathbf{X}_\tau, \mathbf{Z}} [\|\mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau) - \tilde{\mathbf{s}}(\mathbf{X}_\tau, \mathbf{Z}, \tau)\|_2^2] d\tau \quad (15)$$

where $D_F(\mathbf{X}_1 \| \tilde{\mathbf{X}}_1) = \mathbb{E}_{\mathbf{X}_\tau, \mathbf{Z}} [\|\mathbf{s}(\mathbf{X}_\tau, \mathbf{Z}, \tau) - \mathbf{s}(\tilde{\mathbf{X}}_\tau, \mathbf{Z}, \tau)\|_2^2]$ is the Fisher divergence between posteriors $p(\mathbf{X}_1 | \mathbf{X}_\tau, \mathbf{Z})$ and $p(\tilde{\mathbf{X}}_1 | \tilde{\mathbf{X}}_\tau, \mathbf{Z}, \tau)$ [3, 4]. The proposed flow-matching-based similarity measure $\mathcal{S}_{\mathbf{Z},p}$ for $p = 2$ thus serves as a time-integrated, weighted Fisher divergence. The Fisher divergence [3, 4] serves as the fundamental metric for distribution comparison in diffusion-based systems, extending the principles of information geometry into the realm of high-dimensional generative modeling.

The relationship between Fisher divergence and entropy [5] is best characterized as a functional link between the "geometry" of a probability distribution and its informational state. While Shannon entropy provides a global measure of the average uncertainty or information content in a system, Fisher divergence measures the sensitivity of the log-density to perturbations—effectively quantifying the local curvature of the entropy landscape. Because the rate of entropy production in a diffusive system is exactly equal to the Fisher information, the Fisher divergence provides a rigorous path-based metric for how quickly one distribution evolves toward another. In practical terms, this allows researchers to use the Fisher divergence as a surrogate objective to drive complex, non-Gaussian distributions toward a target state, ensuring that the inversion process captures not just the mean behavior, but the precise structural information required to satisfy the system's entropy balance.

4.3 Synthesis: MSE Decomposition and the Fisher Information Limit

Let us assume that $\mathbf{X}_1 | \mathbf{Z}$ is the true target posterior and $\tilde{\mathbf{X}}_1 | \mathbf{Z}$ the posterior sampled by a given inversion method.

The Mean Squared Error (MSE) is a classic evaluation metric to assess the performance of the inversion algorithm. We can decompose the total error w.r.t. the fundamental information-theoretic limits. Let $\mathbf{e}(\tau) = \mathbf{X}_1 - \tilde{\mathbf{X}}_1$ denote the estimation error at flow stage $\tau \in [0, 1]$. The associated MSE $\text{MSE}(\tau)$ is partitioned as:

$$\text{MSE}(\tau) = \left(\frac{\alpha^2(\tau)}{\beta(\tau)} \right)^2 \mathbb{E}_{\mathbf{X}_\tau, \mathbf{Z}} [\|\mathbf{s}^*(\mathbf{X}_\tau, \mathbf{Z}, \tau) - \mathbf{s}_{\tilde{\mathbf{X}}_1}(\mathbf{X}_\tau, \mathbf{Z}, \tau)\|^2] + \text{Tr}(\mathcal{I}^{-1}(\tau)) \quad (16)$$

where $\sigma^4 \mathbb{E}[\|\mathbf{s}^* - \mathbf{s}_{\tilde{\mathbf{X}}_1}\|^2]$ represents the Fisher Divergence D_F , quantifying the excess error due to suboptimal score estimation (the *Information Deficit*), and $\text{Tr}(\mathcal{I}^{-1}(\tau))$ is the Cramér-Rao Lower Bound (CRB), representing the irreducible path-dependent variance floor determined by the signal-to-noise ratio at time τ in the true distribution $p(\mathbf{X}_1 | \mathbf{X}_\tau, \mathbf{Z})$.

This decomposition reveals that our similarity measure $\mathcal{S}_{\mathbf{Z},2}^*$, as the integrated Fisher Divergence, serves as a precise diagnostic: it quantifies the portion of the total MSE that is attributable to model-specific inaccuracies. An optimal inversion method satisfies $\mathcal{S}_{\mathbf{Z},2}^* \rightarrow 0$, effectively saturating the Cramér-Rao bound and achieving the theoretical minimum error imposed by the Fisher information geometry of the posterior.

We can also emphasize that, at flow stage $\tau = 0$, we retrieve the information-theoretic lower bound of the unbiased minimum-variance estimator [6]. By integrating w.r.t. flow stage $\tau \in [0, 1]$, the considered similarity measure can also be decomposed according to the difference between the averaged MSE and CRB.

References

- [1] Y. Lipman, R. Chen, H. Ben-Hamu, M. Nickel, and Matt Le. Flow Matching for Generative Modeling, February 2023.
- [2] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic Interpolants: A Unifying Framework for Flows and Diffusions, October 2025. arXiv:2303.08797 [cs].
- [3] A. Hyvärinen. Estimation of Non-Normalized Statistical Models by Score Matching. *Journal of Machine Learning Research*, 6(24):695–709, 2005.

- [4] S.-I. Amari and H. Nagaoka. *Methods of Information Geometry*. American Mathematical Soc., 2000. Google-Books-ID: vc2FWS07wLUC.
- [5] A. Dembo, T.M. Cover, and J.A. Thomas. Information theoretic inequalities. *IEEE Transactions on Information Theory*, 37(6):1501–1518, November 1991.
- [6] C.M. Bishop and H. Bishop. *Deep Learning: Foundations and Concepts*. 2025.